

Chapter 3. **GROUPS**

3.1 What is a group?

3.1.1 Cayley tables



Today we will talk about **CAYLEY TABLES**.

The Cayley table of a FINITE group

Let $(G, \cdot)$ be a **finite group**. The operation $\cdot$ of G can be completely determined by giving a table, called the **group table** or **Cayley table** of G . Its rows and columns are labelled by the elements of G , and for a and b in G , the element $a \cdot b$ is the entry of the table in the row corresponding to a , and the column corresponding to b .

↓

$\cdot$	$\dots \quad b \quad \dots$
$\vdots$	$\circlearrowleft$ $a \cdot b$
a	
$\vdots$	

→

Example 3.4. Let us write the **Cayley table of S_3** , the symmetric group of degree 3. Recall that

$$S_3 = \{\varepsilon, (1\ 2\ 3), (1\ 3\ 2), (1\ 2), (1\ 3), (2\ 3)\}.$$

To ease the notation, let $\sigma = (1\ 2\ 3)$, $\tau = (1\ 2)$. We have that

$$\sigma^2 = (1\ 3\ 2), \quad \sigma^3 = \varepsilon = \tau^2, \quad \sigma\tau\sigma = \tau,$$

$$\tau\sigma = (2\ 3), \quad \tau\sigma^2 = (1\ 3).$$

Notice that we can now write S_3 as

$$S_3 = \{\varepsilon, \sigma, \sigma^2, \tau, \tau\sigma, \tau\sigma^2\}.$$

τ, σ generators
of S_3

$$\left. \begin{array}{l} \sigma^3 = \varepsilon = \tau^2 \\ \sigma\tau\sigma = \tau \end{array} \right\} \begin{array}{l} \text{relations} \\ \text{between the generators} \end{array}$$

Cayley table of S_3

		ε	σ	σ^2	τ	$\tau\sigma$	$\tau\sigma^2$
S_3		ε	σ	σ^2	τ	$\tau\sigma$	$\tau\sigma^2$
ε		ε	σ	σ^2	τ	$\tau\sigma$	$\tau\sigma^2$
σ		σ	σ^2	ε	$\tau\sigma^2$	τ	$\tau\sigma$
σ^2		σ^2	ε	σ	$\tau\sigma$	$\tau\sigma^2$	τ
τ		τ	$\tau\sigma$	$\tau\sigma^2$	ε	σ	σ^2
$\tau\sigma$		$\tau\sigma$	$\tau\sigma^2$	τ	σ^2	ε	σ
$\tau\sigma^2$		$\tau\sigma^2$	τ	$\tau\sigma$	σ	σ^2	ε

$\checkmark \quad \underline{\sigma\tau = \tau\sigma^2} \quad ; \quad \underline{\sigma^2\tau = \tau\sigma} \quad \checkmark$

$\sigma\tau\sigma = \tau \Rightarrow \sigma^3\tau\sigma = \sigma^2\tau \Rightarrow \underline{\tau\sigma = \sigma^2\tau}$

$\sigma\tau\sigma = \tau \Rightarrow \sigma\tau\sigma^3 = \tau\sigma^2 \Rightarrow \underline{\sigma\tau = \tau\sigma^2}$

$\underline{(\tau\sigma^2)(\tau\sigma)} = \tau(\underline{\sigma^2\tau})\sigma = \tau(\tau\sigma)\sigma = \tau^2\sigma^2$

$\underline{\tau^2 = \varepsilon}$

Some properties of G that we can spot by looking at its Cayley table

- G is **Abelian** if and only if its Cayley table is symmetric about its main diagonal.
- $e \in G$ is the **identity element** 1 of G if and only if the e th row and e th column look the same as the labels.
- the **inverse of a is b** if and only if the entry in row a and column b ($a \cdot b$) and the entry in row b and column a ($b \cdot a$) turn out to be 1.

↓

$\mathbb{Z}_2 \times \mathbb{Z}_2$	$(\bar{0}, \bar{0})$	$(\bar{0}, \bar{1})$	$(\bar{1}, \bar{0})$	$(\bar{1}, \bar{1})$
$(\bar{0}, \bar{0})$	$(\bar{0}, \bar{0})$	$(\bar{0}, \bar{1})$	$(\bar{1}, \bar{0})$	$(\bar{1}, \bar{1})$
$(\bar{0}, \bar{1})$	$(\bar{0}, \bar{1})$	$(\bar{0}, \bar{0})$	$(\bar{1}, \bar{1})$	$(\bar{1}, \bar{0})$
$(\bar{1}, \bar{0})$	$(\bar{1}, \bar{0})$	$(\bar{1}, \bar{1})$	$(\bar{0}, \bar{0})$	$(\bar{0}, \bar{1})$
$(\bar{1}, \bar{1})$	$(\bar{1}, \bar{1})$	$(\bar{1}, \bar{0})$	$(\bar{0}, \bar{1})$	$(\bar{0}, \bar{0})$

$\mathbb{Z}_2 \times \mathbb{Z}_2$ is abelian

↓

S_3	ε	σ	σ^2	τ	$\tau\sigma$	$\tau\sigma^2$
ε	ε	σ	σ^2	τ	$\tau\sigma$	$\tau\sigma^2$
σ	σ	σ^2	ε	$\tau\sigma^2$	τ	$\tau\sigma$
σ^2	σ^2	ε	σ	$\tau\sigma$	$\tau\sigma^2$	τ
τ	τ	$\tau\sigma$	$\tau\sigma^2$	ε	σ	σ^2
$\tau\sigma$	$\tau\sigma$	$\tau\sigma^2$	τ	σ^2	ε	σ
$\tau\sigma^2$	$\tau\sigma^2$	τ	$\tau\sigma$	σ	σ^2	ε

S_3 is not abelian

$\sigma^2 \cdot \sigma = \varepsilon$
 $\sigma \cdot \sigma^2 = \varepsilon$